

TIVE B.P. KHUMALO

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EXPERIENCE

Graduate Research & Teaching Assistant

University of Missouri, Columbia, MO

August 2026 – July 2027

- Working on the development of agentic AI for advanced statistical modeling under the supervision of Brian Keller, Ph.D.
- Grading assignments and assisting graduate students in Multilevel Modeling under the supervision of Brian Keller, Ph.D.

Graduate Research Assistant

University of Missouri, Columbia, MO

July 2024 – July 2026

- Worked in the Comprehensive Statistical Model Evaluation (CoSME) lab with Wes Bonifay, Ph.D., and Sonja D. Winter, Ph.D. on a simulation study on model evaluation, primarily focusing on fit propensity in statistical models.
- Worked also in the Winter Lab with Sonja D. Winter (supervisor) on a Bayesian SEM simulation for single indicator models.

Graduate Research Assistant

University of Missouri, Columbia, MO

August 2024 – February 2025

- Worked on the Supporting Effective Educator Development Grant Program with Amanda Olsen, Ph.D.

Graduate Teaching Assistant

University of Missouri, Columbia, MO

July 2023 – May 2024

- Designing and developing an online section for Quantitative Methods (Statistical Methods) using HTML and R under the supervision of Matt Easter, Ph.D.

Statistical Consultant

University of Missouri, Columbia, MO

July 2025 – Present

- I work as a statistical consultant through Quantitative Statistical Support Initiative, offered by the Office of Research Support in collaboration with the Statistics, Measurement and Evaluation in Education department.
- Provide research design, statistical analysis, and quantitative consulting services to the College of Education and Human Development (CEHD) faculty, postdoctoral researchers, and graduate students.

Science Educator

Teaching Services Commission of Eswatini, Mbabane, Eswatini

March 2020 – March 2022

EDUCATION

University of Missouri, Columbia, MO

August 2024 – May 2028

Doctor of Philosophy (Ph.D.) in Educational, School & Counseling Psychology,
Program Area: Statistics, Measurement, and Evaluation in Education (Emphasis in
Statistics and Measurement)

University of Missouri, Columbia, MO

August 2022 – May 2024

Master of Arts (M.A.) in Educational, School & Counseling Psychology, Program
Area: Statistics, Measurement and Evaluation in Education (Emphasis in Statistics and
Measurement)

University of Eswatini, Kwaluseni, Eswatini

August 2015 – May 2019

Bachelor of Science (B.Sc.), Physics and Mathematics

GRANT FUNDING

Completed

1. Innovative, Translational, and User-Friendly tools for Comprehensive Statistical Model Evaluation (05/01/2024-08/31/2025). Grant awarded by the US Department of Education, Institute for Educational Sciences. PI: Wes Bonifay (University of Missouri). (\$900,000). **Role: Graduate Research Assistant**

2. Increasing Teacher's Capacity for Integrating Mathematics, Social-Emotional Learning, and Equity (Maths + SEL + E) (10/01/2022 – 05/11/2025). Funded by the U.S. Department of Education, Office of Elementary and Secondary Education, Supporting Effective Educator Development Program (PR # S423A220084) (\$ 5, 645, 548). **Role: Graduate Research Assistant**

SKILLS

- Quantitative Analysis
- Measurement (Psychometrics)
- Research
- Advanced Statistical Modeling and Computation (Machine Learning)
- Bayesian Modeling
- Spatial Statistics Modeling
- Latent Variable Modeling
- R, Stan, Python, Blimp, C++, NIMBLE, JAGS, Fortran, and SQL

AWARDS

- Fulbright Scholarship Grantee (2022-2024).
- Paul C. Polmantier Memorial Scholarship Grantee (2025-2026)
- Edward G. and Elizabeth Bower Family Scholarship (2026 – 2027)

CERTIFICATIONS AND LICENSES

Utrecht University

February 2025

- Beyond Null-Hypothesis Testing: Bayesian Informative Hypothesis Evaluation.

PUBLICATIONS

Peer-Reviewed Articles

1. Bergin, C., Roos, J., B. Khumalo, Brockmeier, L., Baron O., and Vogts, A. (2026). Development and Validation of the Prosocial Behavior in School Scale. *Psychology in the Schools*, 63, 1001-1019. <https://doi.org/10.1002/pits.70144>

PAPERS IN PREPARATION

1. Khumalo, T.B.P., Winter, S.D., Bonifay, W. Fitting propensity analysis in structural equation modeling: Design considerations and practical guidelines . [Manuscript Submitted]